

Ather Energy: Driving the future of mobility in India with BigQuery and Cloud IoT Core

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ABOUT Ather Energy

Ather Energy built intelligent electric scooters featuring route optimization and predictive maintenance, to improve lives in India's dynamic economy with Cloud IoT Core and BigQuery.



About Ather Energy

Ather Energy is a Bengaluru-based startup that launched India's first intelligent electric scooter, providing mobility solutions to people from a range of backgrounds. Founded in 2013 by two university friends, it raised \$51 million in Series C funding to evolve to its next phase via machine learning. The company believes that intelligent vehicles will revolutionize the commute experience.

Industries: Manufacturing

Google Cloud results

- Empowers a small engineering team to invent a unique IoT mobility system
- Facilitates platform updates every month instead of twice per year
- Enables deployment of 43 IoT sensors on each vehicle for granular data on driver experience
- Delivers data insights that keep down costs

Instant crunch thousands of scooters

Location: India

(<https://cloud.google.com/ai-platform/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/maps-platform/>)

Cloud IoT Core (<https://cloud.google.com/iot-core/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

AI Platform (<https://cloud.google.com/ai-platform/>)

Google Maps Platform

(<https://cloud.google.com/maps-platform/>)

**for both the
company
and the
consumer**

Swapnil Jain and Tarun Mehta became fascinated by mobility solutions for India's fast-growing economy while studying in Chennai, a cosmopolitan city on the Bay of Bengal. At IIT Madras, India's top-ranked engineering university, the friends immersed themselves in experimental vehicle projects that helped them win elite auto industry jobs. But they soon felt drawn to a bigger mission than climbing the corporate ladder.

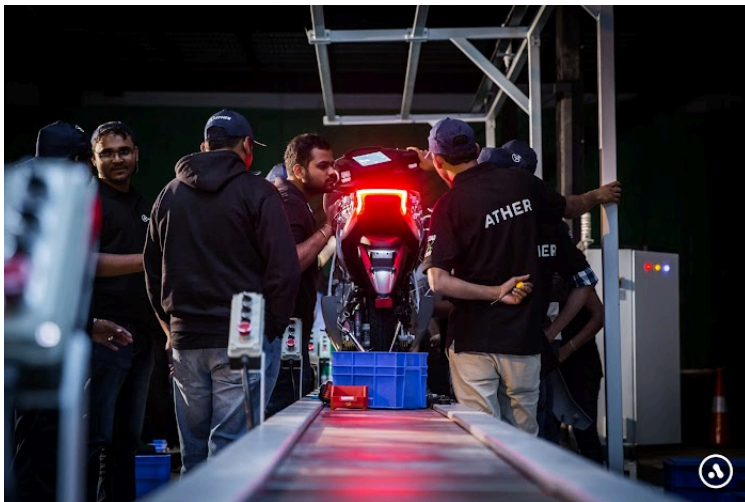
"We were itching to do something different," says Swapnil. "We wanted to explore the possibilities of electric vehicles and how they can be developed to empower India."

Swapnil and Tarun launched Ather Energy (<https://www.atherenergy.com/>) from a belief that electric scooters could help drive the growth narrative of a nation of 1.3 billion on the path to middle-income status. But they knew that could only be part of the picture: "As a startup, we have to build something that's redefining the segment," says Swapnil.

For the partners, segment redefinition meant inventing an intelligent electric scooter, powered by AI and IoT, that would enable people to navigate the chaotic maze of Indian life. Yet with a small team of engineers and limited resources, they needed more

than innovation smarts to succeed. They linked their big idea to Google Cloud (<https://cloud.google.com/>), confident that Cloud IoT Core (<https://cloud.google.com/iot-core>) and BigQuery (<https://cloud.google.com/bigquery/>) would give them freedom to buckle down to the task of creating unique mobility solutions.

“That was the entry into Google Cloud,” says Swapnil, “and then we just went with everything else on top of it, since the solutions can be easily deployed and scaled up.”



Ather Energy Factory

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—Swapnil Jain, Co-
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Connected mobility for a country on the move

Driving in India can be tricky, with roads cratered with potholes, streets with no names, chronic traffic congestion, confusing city layouts, and free-for-all driving. Swapnil and Tarun had a vision to tackle such problems with an electric scooter that constantly evolves based on road data, driving habits, and vehicle performance.

But Ather Energy's team of 100 employees had more on its to-do list than building intelligent applications. They also needed to design a brand new lithium-ion battery capable of 75km distances, develop a motor powerful enough to endure Indian roads, and create a touchscreen dashboard that would survive intense heat, dust storms, and road vibrations. With such a set of challenges, Ather Energy didn't have time to spare on developing data analytics or maintaining IT infrastructure. By turning to Google Cloud, Ather Energy was able to focus on creating a machine for India's future, while entrusting the big data processing to Cloud IoT Core and BigQuery.

"We wanted a solution that could be put on the market by a really small team," says Swapnil. "We have a lot of data that needs to be analyzed at a very rapid pace so BigQuery was the most obvious choice. With Google Cloud, I have a world-class product and don't have to build a team for it. It means I can go back to focusing on my core task, which is the application."

The Ather 450, Ather Energy's premium product, derives uniqueness from 43 IoT sensors fitted into each scooter, enabling it to obtain granular information on every aspect of the driver experience. The deployment of so many sensors, says Swapnil, was made possible by the virtually unlimited computing power of Google Cloud.

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Ather 450

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Road to success with superior data solutions

The synergies between Ather Energy and Cloud IoT solutions resulted in a unique mobility platform for India and beyond. With [Google Maps Platform](https://cloud.google.com/maps-platform/) (https://cloud.google.com/maps-platform/)-enabled navigation as a core feature, Ather Energy’s platform capabilities are amplified. The Ather 450 performs predictive maintenance to fix problems before they happen, structures electricity use for big cost savings, analyzes traffic and road conditions for optimal route options, and alerts riders to unoccupied electricity stations in the Ather Grid, the startup’s charging infrastructure.

Meanwhile, the aggregated data from all scooters on the road is providing key intelligence for Swapnil and Tarun to develop future models based on customer habits and requirements. The company, for example, can read exactly how far customers are riding every day to inform decisions regarding driving range in product development.

“We’re churning out a few hundred megabytes of data every day, on every vehicle,” says Swapnil. “With that sort of data, and a few thousand vehicles on the road, we’re doing things which would have been impossible without Google Cloud.”

Critically, Google Cloud software and architecture solutions have enabled Ather Energy’s team to deploy microservices in real time at a constant rate so it can deliver more frequent monthly platform updates that help capitalize on what a connected vehicle can do. Innovative updates such as "guide-me-home" lights and real-time ride/energy efficient indicators help bolster Ather Energy’s competitive position.

“One of the big things about having connected vehicles is that you don't want to roll out applications in a six-month time frame,” explains Swapnil. “You want to be rolling out an update every month. And that means our focus has to be on application development and not on infrastructure and maintenance of any sort.”

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Curbing costs to bring intelligent mobility to everyday life

Ather Energy has been met with rising demand from people with various backgrounds, not just the tech-savvy niche it initially expected. The topline Ather scooter costs roughly US\$1,800, a big outlay in a country with per capita income at purchasing power parity of about US\$7,760. Yet consumers are grasping how benefits such as structured electricity plans, predictive maintenance, and route optimization eventually save them money.

“Word of mouth is spreading, with people realizing more and more how ownership costs are down, and experiences are a lot better,” says Swapnil.

Meanwhile, data insights that Ather Energy gains from BigQuery deliver cost savings to the company itself, which can be passed onto the consumer: “We can make a small tweak into some of the firmware and fix issues with no cost impact on the customer as well as for us,” says Swapnil.

Machine learning for growth without limits

In the next step in its partnership with Google Cloud, Ather Energy, which now has 650 employees, plans to turn to Google Cloud machine learning solutions such as [AI Platform](https://cloud.google.com/ai-platform/) (https://cloud.google.com/ai-platform/) to add self-correcting capabilities to their intelligent vehicles.

Thousands of connected Ather scooters across a dynamic Indian landscape are feeding a treasure trove of data back home. Google Cloud machine learning tools will process that information to enable the vehicles to start learning on their own and avoid situations such as potholes and traffic jams.

“We want to start building machine learning solutions with Google Cloud as rapidly as possible,” says Swapnil. “The next big area we’ll focus on are features that will improve the ownership and daily ride experience. Our upcoming OTAs will include features like predictive maintenance, improved

range prediction, and adding first to category abilities like rider profiling.”

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business.

